1

Combinational Logic Circuits

1.1 Introduction

The aim in this experiment is to find the expression with the lowest cost for combinational logic circuits and implement them.

1.2 Preliminary

1. Find all prime implicants of the function F below.
   a) using Karnaugh diagram
   b) using Quine-McCluskey method
   $$F(a,b,c,d) = \cup_1(0,1,5,14,15) + \cup_\phi(7,8,10,13)$$
   Also create the prime implicant chart of the function F and find the expression with the lowest cost. Cost criteria is 2 units for each variable and 1 unit for each complement. Draw the lowest cost expression using AND, OR, and NOT gates.
2. Design and draw the same function using only NAND and NOT gates.
3. Design and draw the same function using a single 8:1 multiplexer and NOT gates.
4. Design and draw the function below using a single 3:8 decoder, OR, and NOT gates.
   $$F_1(a,b,c) = a' \cdot c' + b \cdot c$$
   $$F_2(a,b,c) = a' \cdot b' \cdot c' + a \cdot b$$

1.3 Equipments and Integrated Circuits (ICs)

Following equipment and ICs are going to be used in the experiment.

- C.A.D.E.T. (**C**omplete **A**nalogue **D**igital **E**lectronic **T**rainer)
- 74000 series ICs
  - $74xx00$ - Quadruple 2-input Positive NAND Gates
  - $74xx04$ - Hex Inverters
  - $74xx08$ - Quadruple 2-input Positive AND Gates
  - $74xx10$ - Triple 3-input Positive NAND Gates
  - $74xx11$ - Triple 3-input Positive AND Gates
  - $74xx27$ - Triple 3-input Positive NOR Gates
  - $74xx32$ - Quadruple 2-input Positive OR Gates

- $74xx138$ - 3:8 Decoder
- $74xx151$ - 8:1 Multiplexer

Fundamental information (function tables and pin configurations) of the ICs listed above are given in the Appendix A. You should also examine the data-sheets in order to acquire further information about these ICs.

1.4 Experiment

1.4.1 Experiment - Part 1

Build the circuit you have designed in Preliminary Question 1 using the ICs. Fill up a truth table to evaluate your implementation.

1.4.2 Experiment - Part 2

Build the circuit you have designed in Preliminary Question 2 using the ICs. Fill up a truth table to evaluate your implementation.

1.4.3 Experiment - Part 3

Build the circuit you have designed in Preliminary Question 3 using the ICs. Fill up a truth table to evaluate your implementation.

1.4.4 Experiment - Part 4

Build the circuit you have designed in Preliminary Question 4[1] using the ICs. Fill up a truth table to evaluate your implementation.

1.5 Report

Prepare your report by using the guidelines and the report template which are posted on Ninova e-Learning System. Your report should also include the following materials:

- Circuits diagrams of the expressions which were implemented during this experiment.
- Function tables (or truth tables) of the implemented expressions.
- Discuss the output of your circuits for undetermined (ϕ) inputs. Why did you get these outputs?

During the experiment, please do not forget to take notes about the critical points of the implementations in order to write a proper report for the experiment. Additionally, if there were any complications which affect your performance during the experiment, please also indicate these difficulties in your report.

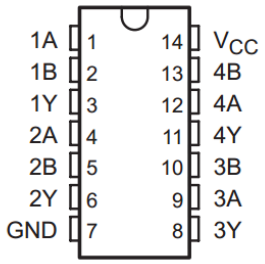
[1] Notice that, the given decoder IC (**74xx138**) in the experiment works in **negative logic** mode. Thus, you need to re-consider your design for the output logic accordingly.

A

Appendix - Data Sheets

A.1 7400 - Quadruple 2-input Positive-NAND Gates

Pin Configuration

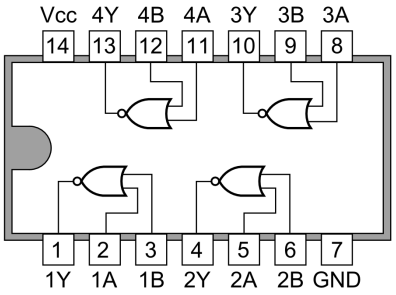


Function Table

INPUTS		OUTPUT
A	B	Y
H	H	L
L	X	H
X	L	H

A.2 7402 - Quad 2-input Positive-NOR Gates

Pin Configuration



Function Table

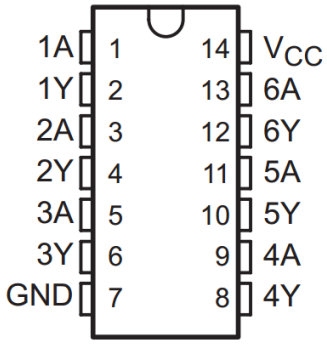
$Y = \overline{A + B}$

Inputs		Output
A	B	Y
L	L	H
L	H	L
H	L	L
H	H	L

H = HIGH Logic Level
L = LOW Logic Level

A.3 7404 - Hex Inverters

Pin Configuration

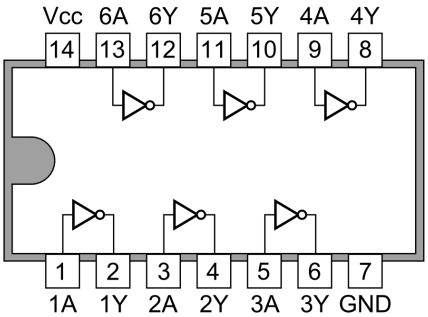


Function Table

INPUT A	OUTPUT Y
H	L
L	H

A.4 7405 - Hex Inverters with Open-Collector Outputs

Pin Configuration



Function Table

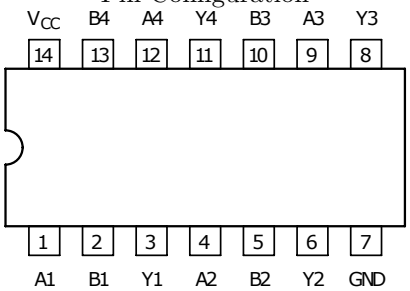
$Y = \overline{A}$

Input A	Output Y
L	H
H	L

H = HIGH Logic Level
L = LOW Logic Level

A.5 7408 - Quadruple 2-input Positive-AND Gates

Pin Configuration

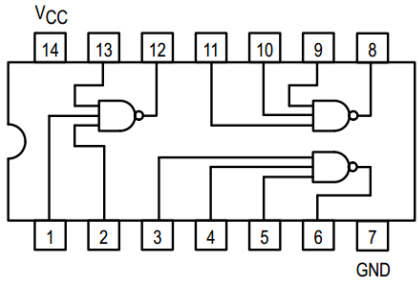


Function Table

Inputs A	Inputs B	Output Y
L	L	L
L	H	L
H	L	L
H	H	H

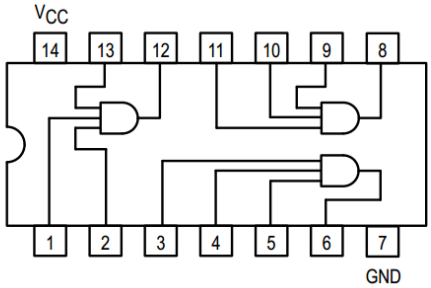
A.6 7410 - Triple 3-input NAND Gates

Pin Configuration



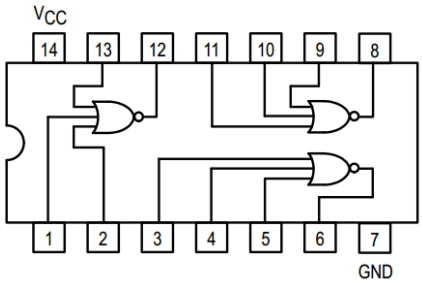
A.7 7411 - Triple 3-input AND Gates

Pin Configuration



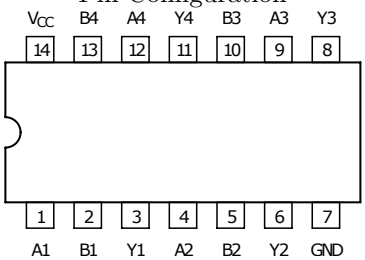
A.8 7427 - Triple 3-input NOR Gates

Pin Configuration



A.9 7432 - Quadruple 2-input Positive-OR Gates

Pin Configuration

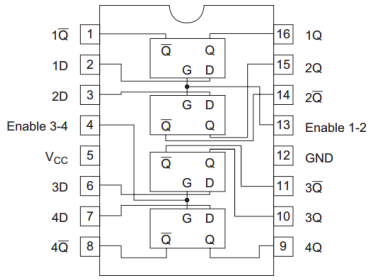


Function Table

Inputs		Output
A	**B**	**Y**
L	L	L
L	H	H
H	L	H
H	H	H

A.10 7475 - Quadruple Bistable Latches

Pin Configuration



Function Table

Inputs		Outputs	
D	G	Q	$\overline{Q}$
L	H	L	H
H	H	H	L
X	L	Q_0	$\overline{Q}_0$

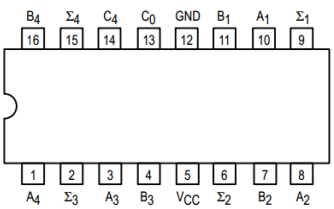
H; high level, L; low level, X; irrelevant

Q_0; level of Q before the indicated steady-state input conditions were established.

$\overline{Q}_0$; complement of Q_0 or level of $\overline{Q}_0$ before the indicated steady-state input conditions were established.

A.11 7483 - 4-Bit Binary Full Adder with Fast Carry

Pin Configuration



Function Table

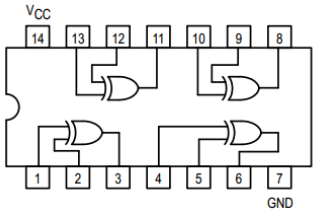
FUNCTIONAL TRUTH TABLE

C (n–1)	A_n	B_n	Σ_n	C_n
L	L	L	L	L
L	L	H	H	L
L	H	L	H	L
L	H	H	L	H
H	L	L	H	L
H	L	H	L	H
H	H	L	L	H
H	H	H	H	H

C_1 — C_3 are generated internally
C_0 — is an external input
C_4 — is an output generated internally

A.12 7486 - Quad 2-Input Exclusive Or Gate

Pin Configuration



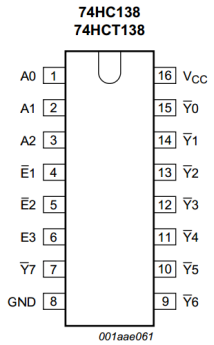
Function Table

TRUTH TABLE

IN		OUT
A	B	Z
L	L	L
L	H	H
H	L	H
H	H	L

A.13 74138 - 3-to-8 decoder/demultiplexer

Pin Configuration



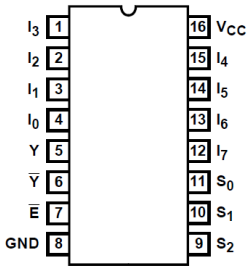
Function Table

Control			Input			Output							
$\overline{E}1$	$\overline{E}2$	E3	A2	A1	A0	$\overline{Y}7$	$\overline{Y}6$	$\overline{Y}5$	$\overline{Y}4$	$\overline{Y}3$	$\overline{Y}2$	$\overline{Y}1$	$\overline{Y}0$
H	X	X	X	X	X	H	H	H	H	H	H	H	H
X	H	X											
X	X	L											
L	L	H	L	L	L	H	H	H	H	H	H	H	L
			L	L	H	H	H	H	H	H	H	L	H
			L	H	L	H	H	H	H	H	L	H	H
			L	H	H	H	H	H	H	L	H	H	H
			H	L	L	H	H	H	L	H	H	H	H
			H	L	H	H	H	L	H	H	H	H	H
			H	H	L	H	L	H	H	H	H	H	H
			H	H	H	L	H	H	H	H	H	H	H

[1] H = HIGH voltage level;
L = LOW voltage level;
X = don't care.

A.14 74151 - 8-Input Multiplexer

Pin Configuration

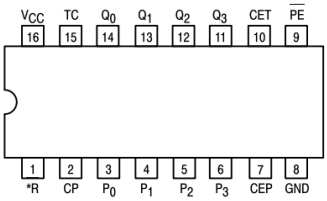


Function Table

INPUTS												OUTPUTS	
$\overline{E}$	S_2	S_1	S_0	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7	$\overline{Y}$	Y
H	X	X	X	X	X	X	X	X	X	X	X	H	L
L	L	L	L	L	X	X	X	X	X	X	X	H	L
L	L	L	L	H	X	X	X	X	X	X	X	L	H
L	L	L	H	X	L	X	X	X	X	X	X	H	L
L	L	L	H	X	H	X	X	X	X	X	X	L	H
L	L	H	L	X	X	L	X	X	X	X	X	H	L
L	L	H	L	X	X	H	X	X	X	X	X	L	H
L	L	H	H	X	X	X	L	X	X	X	X	H	L
L	L	H	H	X	X	X	H	X	X	X	X	L	H
L	H	L	L	X	X	X	X	L	X	X	X	H	L
L	H	L	L	X	X	X	X	H	X	X	X	L	H
L	H	L	H	X	X	X	X	X	L	X	X	H	L
L	H	L	H	X	X	X	X	X	H	X	X	L	H
L	H	H	L	X	X	X	X	X	X	L	X	H	L
L	H	H	L	X	X	X	X	X	X	H	X	L	H
L	H	H	H	X	X	X	X	X	X	X	L	H	L
L	H	H	H	X	X	X	X	X	X	X	H	L	H

A.15 74161 - BCD Decade Counter / 4-Bit Binary Counter

Pin Configuration

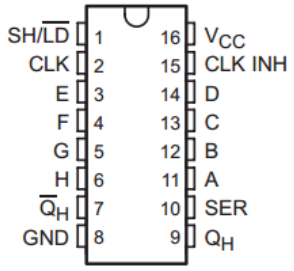


Function Table

*SR	PE	CET	CEP	Action on the Rising Clock Edge (⎍)
L	X	X	X	RESET (Clear)
H	L	X	X	LOAD (P_n Q_n)
H	H	H	H	COUNT (Increment)
H	H	L	X	NO CHANGE (Hold)
H	H	X	L	NO CHANGE (Hold)

A.16 74165 - 8-Bit Parallel-Load Shift Register

Pin Configuration



Function Table

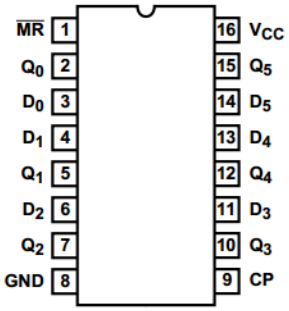
FUNCTION TABLE

INPUTS			FUNCTION
SH/$\overline{LD}$	CLK	CLK INH	
L	X	X	Parallel load
H	H	X	No change
H	X	H	No change
H	L	↑	Shift†
H	↑	L	Shift†

† Shift = content of each internal register shifts toward serial output Q_H. Data at SER is shifted into the first register.

A.17 74174 - Hex D-Type Flip-Flop with Reset

Pin Configuration



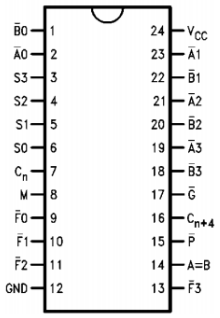
Function Table

INPUTS			OUTPUT
RESET ($\overline{MR}$)	CLOCK CP	DATA D_n	Q_n
L	X	X	L
H	↑	H	H
H	↑	L	L
H	L	X	Q_0

H = High Voltage Level, L = Low Voltage Level, X = Irrelevant, ↑ = Transition from Low to High Level, Q_0 = Level Before the Indicated Steady-State Input Conditions Were Established

A.18 74181 - 4-Bit Arithmetic Logic Unit

Pin Configuration & Function Table



Pin Names	Description
$\overline{A}0$–$\overline{A}3$	Operand Inputs (Active LOW)
$\overline{B}0$–$\overline{B}3$	Operand Inputs (Active LOW)
S0–S3	Function Select Inputs
M	Mode Control Input
C_n	Carry Input
$\overline{F}0$–$\overline{F}3$	Function Outputs (Active LOW)
A = B	Comparator Output
$\overline{G}$	Carry Generate Output (Active LOW)
$\overline{P}$	Carry Propagate Output (Active LOW)
C_{n+4}	Carry Output

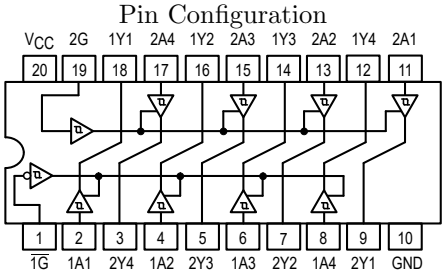
Mode Select Inputs				Active LOW Operands & F_n Outputs		Active HIGH Operands & F_n Outputs	
				Logic	Arithmetic (Note 2)	Logic	Arithmetic (Note 2)
S3	S2	S1	S0	(M = H)	(M = L) (C_n = L)	(M = H)	(M = L) (C_n = H)
L	L	L	L	$\overline{A}$	A minus 1	$\overline{A}$	A
L	L	L	H	$\overline{AB}$	AB minus 1	$\overline{A}+\overline{B}$	A + B
L	L	H	L	$\overline{A}+\overline{B}$	$A\overline{B}$ minus 1	$\overline{A}B$	$A+\overline{B}$
L	L	H	H	Logic 1	minus 1	Logic 0	minus 1
L	H	L	L	$\overline{A}+\overline{B}$	A plus $(A+\overline{B})$	$\overline{AB}$	A plus $A\overline{B}$
L	H	L	H	$\overline{B}$	AB plus $(A+\overline{B})$	$\overline{B}$	$(A+B)$ plus $A\overline{B}$
L	H	H	L	$\overline{A}\oplus\overline{B}$	A minus B minus 1	$A\oplus B$	A minus B minus 1
L	H	H	H	$A+\overline{B}$	$A+\overline{B}$	$A\overline{B}$	AB minus 1
H	L	L	L	$\overline{A}B$	A plus (A + B)	$\overline{A}+B$	A plus AB
H	L	L	H	$A\oplus B$	A plus B	$\overline{A}\oplus\overline{B}$	A plus B
H	L	H	L	B	$A\overline{B}$ plus (A + B)	B	$(A+\overline{B})$ plus AB
H	L	H	H	A + B	A + B	AB	AB minus 1
H	H	L	L	Logic 0	A plus A (Note 1)	Logic 1	A plus A (Note 1)
H	H	L	H	$A\overline{B}$	AB plus A	$A+\overline{B}$	$(A+B)$ plus A
H	H	H	L	AB	$A\overline{B}$ minus A	A + B	$(A+\overline{B})$ plus A
H	H	H	H	A	A	A	A minus 1

Note 1: Each bit is shifted to the next most significant position.

Note 2: Arithmetic operations expressed in 2s complement notation.

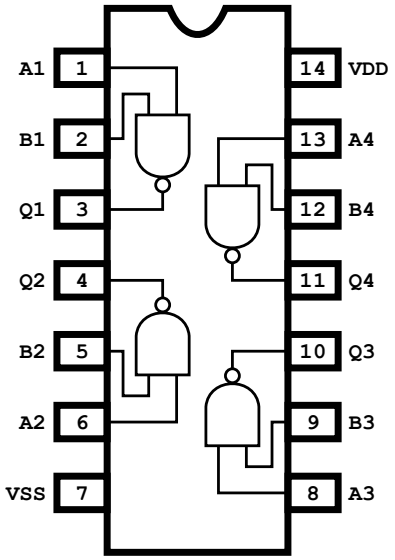
A.19 74241 - Octal 3-State Buffer/ Line Driver/ Line Receiver

Pin Configuration



Function Table

Inputs				Outputs	
G	$\overline{G}$	1A	2A	1Y	2Y
X	L	L	X	L	
X	L	H	X	H	
X	H	X	X	Z	
H	X	X	L		L
H	X	X	H		H
L	X	X	X		Z

A.20 4011 - Quad 2-Input NAND Gate

Pin Configuration